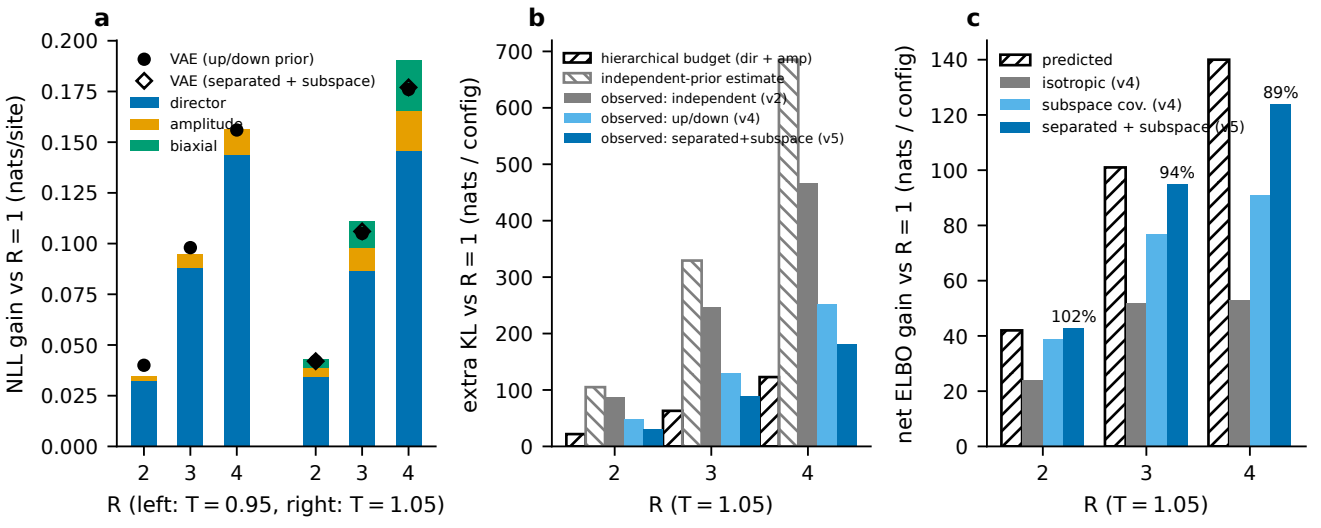
a
NLL gain vs R = 1 (nats/site)
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
VAE (up/down prior)
VAE (separated + subspace)
director
amplitude
biaxial
2
3
4
2
3
4
R (left: T = 0.95, right: T = 1.05)
b
extra KL vs R = 1 (nats / config)
700
600
500
400
300
200
100
0
hierarchical budget (dir + amp)
independent-prior estimate
observed: independent (v2)
observed: up/down (v4)
observed: separated+subspace (v5)
2
3
4
R (T = 1.05)
c
net ELBO gain vs R = 1 (nats / config)
140
120
100
80
60
40
20
0
predicted
isotropic (v4)
subspace cov. (v4)
separated + subspace (v5)
102%
94%
89%
2
3
4
R (T = 1.05)